

I. Heat transfer

Symbology

Symbol	Quantity	Unit
λ, k	Thermal conductivity	W/(m · K)
$\dot{q}$	Heat flux	W/m ²
$\dot{Q}$	Heat flow	W
α	Heat transfer coefficient	W/(m ² · K)
a	Thermal diffusivity	m ² /s
ν	Kinematic viscosity	m ² /s

1. Types of heat transfer

1.1 Heat conduction

Occurs with all materials when there is a temperature gradient.

Steady-state conduction (1st Fourier's law):

$$\dot{q} = -\lambda \frac{dT}{dx}$$

Transient conduction (2nd Fourier's law 1D):

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \frac{\partial^2 T}{\partial x^2}$$

1.2 Convective heat transfer

This occurs in moved fluids.

Newton's approach:

$$\dot{q} = \alpha (T_1 - T_2) = \frac{\lambda_{\text{fluid}}}{\delta_{\text{th}}} (T_1 - T_2)$$

1.3 Radiation

Bodies with a temperature above 0 K emit thermal radiation.

2. Heat conduction and conductivity

2.1 Heat flux

$$\dot{q} = -\lambda \frac{dT}{dx} = k \Delta T = \frac{\lambda}{\delta} \Delta T$$

Procedure:

$$\vec{q} \int_0^\delta dx = -\lambda \int_{T_1}^{T_2} dT \Rightarrow \dot{q} = \frac{\lambda}{\delta} (T_1 - T_2)$$

2.2 Heat flow

$$\dot{Q} = \frac{\Delta Q}{\Delta t} = \dot{q} A = k A \Delta T = \frac{\lambda}{\delta} A \Delta T$$

2.3 Thermal conduction k^{-1} and Thermal conductivity k

$$k_{\text{tot}}^{-1} = \sum_{i=1}^n \frac{\delta_i}{\lambda_i} \Rightarrow k_{\text{tot}} = \frac{1}{k_{\text{tot}}^{-1}}$$

3. Heat conductivity through a hollow cylinder

$$\dot{Q} = -\lambda A \frac{dT}{dr} = -\lambda 2\pi r l \frac{dT}{dr}$$

$$\dot{Q} = 2\pi l \cdot \frac{T_0 - T_1}{\frac{1}{\lambda} \ln \frac{r_1}{r_0}} = 2\pi r_0 l k \Delta T$$

With multilayered walls:

$$\dot{Q} = 2\pi l \cdot \frac{T_0 - T_n}{\sum_{i=1}^n \frac{1}{\lambda_i} \ln \frac{r_i}{r_{i-1}}} = \boxed{2\pi r_0 l k_0 (T_0 - T_n)}$$

3.1 Thermal conductivity k

$$k_0 = \frac{1}{r_0 \left(\left(\frac{1}{\lambda_1} \ln \frac{r_1}{r_0} \right) + \left(\frac{1}{\lambda_2} \ln \frac{r_2}{r_1} \right) + \dots + \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}} \right) \right)}$$

$$k_0^{-1} = r_0 \sum_n \left(\frac{1}{\lambda_n} \ln \frac{r_n}{r_{n-1}} \right)$$

3.2 Thin-walled pipe approximated as a flat wall

$$\dot{Q}_T = \lambda A \frac{T_0 - T_1}{\delta} \quad ; \quad A = 2\pi l \left(r_0 + \frac{\delta}{2} \right)$$

$$\frac{\dot{Q}_T}{\dot{Q}_W} = \frac{\delta}{\left(r_0 + \frac{\delta}{2} \right) \ln \left(1 + \frac{\delta}{r_0} \right)} = \frac{\delta}{r_0} \frac{1}{\left(1 + \frac{\delta}{2r_0} \right) \ln \left(1 + \frac{\delta}{r_0} \right)}$$

4. Convective heat transfer

4.1 Reynold's number

It compares inertial to viscous forces.

$$\boxed{Re = \rho c \frac{L_{\text{char}}}{\eta} = c \frac{L_{\text{char}}}{\nu}} \quad \begin{array}{l} Re < 2300: \text{Laminar flow;} \\ Re \approx 2300: \text{Transition zone;} \\ Re > 2300: \text{Turbulent flow} \end{array}$$

4.2 Prandtl number

It compares momentum diffusivity to thermal diffusivity. Sets velocity's relative thickness and thermal boundary layers.

$$Pr = \frac{\nu}{a} = \frac{\delta_{\text{fluid}}}{\delta_{\text{th}}} = \frac{\eta c_p}{\lambda} \quad ; \quad a = \frac{\lambda}{\rho c_p} \quad ; \quad \nu = \frac{\eta}{\rho}$$

4.3 Nusselt number

Measures convective heat transfer relative to pure conduction across a boundary layer and the heat transfer coefficient α .

$$Nu = \frac{\alpha \cdot L_{\text{char}}}{\lambda_{\text{fl}}} = \dot{q} \frac{L_{\text{char}}}{\lambda \Delta T_{\text{char}}}$$

$$\alpha = \frac{Nu \cdot \lambda}{L_{\text{char}}}$$

4.3.1 Forced convection

$$\boxed{Nu = c \cdot Re^m \cdot Pr^n}$$

where c is a constant.

For laminar flow over a flat plate:

$$Nu = 0.664 \cdot \sqrt{Re} \cdot \sqrt[3]{Pr}$$

For a flow in the transition zone (Tz):

$$Nu_{\text{Tz}} = \sqrt{Nu_{\text{lam}}^2 + Nu_{\text{turb}}^2}$$

4.4 Grashof number

It compares buoyancy to viscous forces.

$$Gr = g \beta \Delta T \frac{L^3}{\nu^2} \quad ; \quad g = 9.81 \text{ N/kg}$$

$$\beta \Delta T = \frac{\rho_\infty - \rho_0}{\rho_\infty} = \frac{\Delta \rho}{\rho_\infty}$$

$$Gr = g \frac{\rho_\infty - \rho_0}{\rho_\infty} \frac{L^3}{\nu^2}$$

4.4.1 Natural convection

Peclet number

It compares advective to diffusive heat transport, i.e., the strength of convection relative to thermal diffusion.

$$Pe = Re Pr$$

Rayleigh number

It combines buoyancy driving and thermal diffusion, and it measures the intensity of natural convection.

$$Ra = Gr Pr$$

5. Heat exchanger

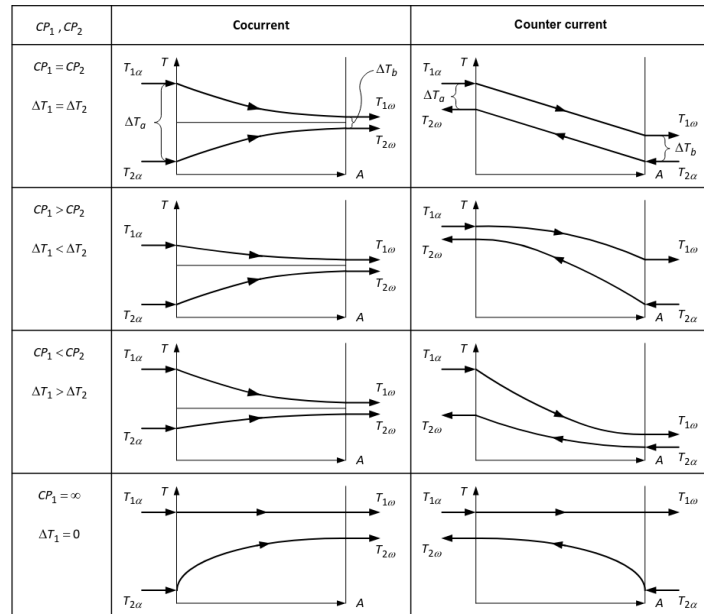
5.1 Energy balance

$$\dot{Q} = \dot{m} c_p \Delta \vartheta = C P \Delta \vartheta$$

$$\dot{Q}_1 = \dot{Q}_2$$

$$\dot{m}_1 c_{p1} \Delta \vartheta_1 = \dot{m}_2 c_{p2} \Delta \vartheta_2 \iff C P_1 \Delta \vartheta_1 = C P_2 \Delta \vartheta_2$$

5.2 Average temperature difference



5.2.1 Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \frac{\Delta T_a}{\Delta T_b}}$$

where:

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

5.2.2 Mean temperature heat flow

$$\dot{Q} = kA\Delta T_m$$

5.2.3 Heat capacity rate ratio (Capacity ration)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$R_1 = \frac{\Delta \vartheta_1}{\Delta \vartheta_2} = \frac{CP_1}{CP_2}$$

5.3 Cocurrent vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{co}\Delta T_{m,co} = kA_{count}\Delta T_{m,count}$$

$$\frac{A_{co}}{A_{count}} = \frac{\Delta T_{m,count}}{\Delta T_{m,co}}$$

5.3.1 If $\dot{Q}_1 \neq \dot{Q}_2$

$$\frac{A_{co}}{A_{count}} = \frac{\dot{Q}_{count}\Delta T_{m,count}}{\dot{Q}_{co}\Delta T_{m,co}}$$

II. The 2nd law of thermodynamics

6. Recap of the 1st law

6.1 Closed systems

$$q_{12} + w_{12} = u_2 - u_1$$

6.2 Open systems

$$q_{12} + w_{12} = h_2 - h_1 + \frac{1}{2}(c_2^2 - c_1^2) + g(z_2 - z_1)$$

$$w_{t12} = \int v dp + \frac{1}{2}(c_2^2 - c_1^2) + g(z_2 - z_1) + j_{12}$$

$$q_{12} + \int v dp + j_{12} = h_2 - h_1 \stackrel{\text{ideal gas}}{=} c_p(T_2 - T_1)$$

7. The 2nd law

7.1 Statements

$$\text{Energy} = \text{Exergy} + \text{Anergy}$$

Processes	Reversible	Irreversible
Energy	Conserved	Reduced → Loss
Entropy	Constant	Increases → Production

7.2 Entropy

Entropy S : extensive state variable [J/K]

Specific entropy s : intensive state variable [J/kgK]

Entropy flow $\dot{S}$: entropy over time [W/K]

8. Entropy balance equation

$$dS = dS_Q + dS_m + dS_{irr}$$

8.1 Entropy balance terms

- Heat transfer across the system boundary:

$$dS_Q = \frac{dQ}{T}$$

- Mass flow across the system boundary:

$$dS_m = \dot{S} = \dot{m} s$$

- Irreversible processes inside the system:

$$\dot{S}_{irr}(t) \begin{cases} dS_{irr} > 0 & \text{irreversible processes} \\ dS_{irr} = 0 & \text{reversible processes} \\ dS_{irr} < 0 & \text{impossible} \end{cases}$$

8.2 Entropy balance equation for closed systems

$$dS = dS_Q + dS_m + dS_{irr}$$

$$dS = \dot{S}_Q dt + \dot{S}_{irr} dt = \left(\frac{\dot{Q}}{T} + \dot{S}_{irr} \right) dt$$

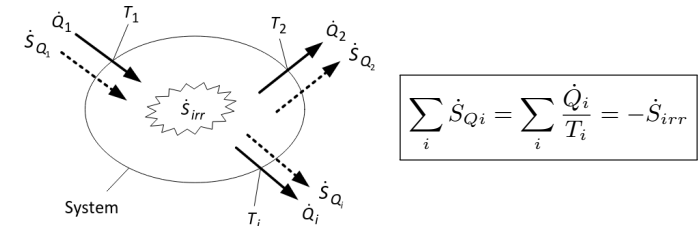
8.2.1 Closed system entropy flow balance equation

$$\frac{dS}{dt} = \sum_i \dot{S}_{Qi} + \dot{S}_{irr} \quad ; \quad \dot{S}_{irr} \geq 0$$

For steady-state case:

$$\dot{S}_{irr} = -\sum_i \dot{S}_{Qi} = -\sum_i \frac{\dot{Q}_i}{T_i} \geq 0$$

8.2.2 Visual representation



8.3 Entropy balance equation for open systems

$$\frac{dS}{dt} = \sum_{\alpha} \dot{m}_{\alpha} s_{\alpha} - \sum_{\omega} \dot{m}_{\omega} s_{\omega} + \sum \dot{S}_Q(t) + \dot{S}_{irr}(t)$$

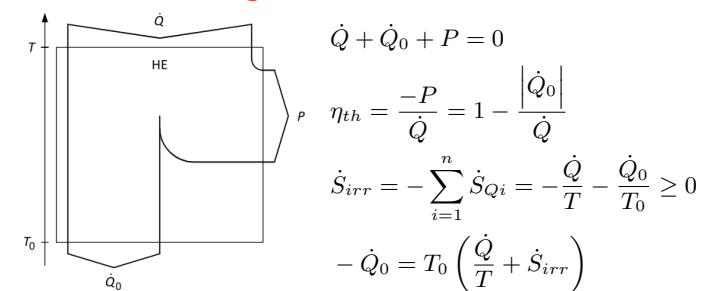
8.3.1 Steady-state flow processes

$$\dot{S}_{irr} = \sum_{\omega} \dot{m}_{\omega} s_{\omega} - \sum_{\alpha} \dot{m}_{\alpha} s_{\alpha} - \sum \dot{S}_Q \geq 0$$

8.3.2 Steady-state and adiabatic flow processes

$$\dot{S}_{irr} = \sum_{\omega} \dot{m}_{\omega} s_{\omega} - \sum_{\alpha} \dot{m}_{\alpha} s_{\alpha} \geq 0$$

9. The thermal enigne



9.1 Carnot efficiency

$$\eta_{th} = 1 - \frac{T_0}{T} - \frac{T_0 \dot{S}_{irr}}{\dot{Q}}$$

$$\eta_{th,max} = \eta_C(T, T_0) = 1 - \frac{T_0}{T}$$

10. Entropy S as a state variable

$$ds = \frac{dq + dj}{T} = \frac{du + p dv}{T} = \frac{dh - v dp}{T}$$

$$T ds = dq + dj \implies T(s_2 - s_1) = q_{12} + j_{12}$$

10.1 Reversible processes

10.1.1 Clausius entropy relation

$$\delta q_{\text{rev}} = T ds \iff \delta Q_{\text{rev}} = T dS$$

10.1.2 Work W and heat Q

$$w_{p, \text{rev}} = \int v dp$$

$$Q_{\text{rev}} = \int T ds$$

10.2 Dissipation j as a state variable

10.2.1 Dissipation

$$w_{t12} = w_{p12} + j_{12} \iff w_{p12} = w_{t12} - j_{12}$$

$$dq_{\text{rev}} + dw_{v, \text{rev}} = du$$

$$T ds = dq + T ds_{\text{irr}} = dq_{\text{rev}} + dj$$

$$\int_1^2 T ds = q_{12} + j_{12}$$

10.2.2 For closed systems

$$ds = \frac{du + p dv}{T} = \frac{d_v dT}{T} + \frac{R_i dv}{v}$$

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R_i \ln \frac{v_2}{v_1}$$

10.2.3 Work change

All the works can be calculated with a temperature change:

$$q_{12} + w_{t12} = h_2 - h_1 + \frac{c_2^2 - c_1^2}{2} + g(z_2 - z_1)$$

$$q_{12} + \int v dp + j_{12} = h_2 - h_1 = c_p (T_2 - T_1)$$

$$\int T ds = q_{12} + j_{12} = \int T ds + \int v dp = h_2 - h_1$$

$$T ds + v dp = h_2 - h_1 \iff T ds - p dv = u_2 - u_1$$

10.3 Reversible state changes with perfect gases

10.3.1 Open system

$$q_{12} + j_{12} + w_{p, \text{rev}} = h_2 - h_1$$

$$\int_1^2 T ds + \int_1^2 v dp = c_p (T_2 - T_1)$$

10.3.2 Closed system

$$q_{12} + j_{12} + w_{p, \text{rev}} = u_2 - u_1$$

$$\int_1^2 T ds - \int_1^2 p dv = c_v (T_2 - T_1)$$

10.4 Polytropic state changes

$$T ds = dq + dj = du + p dv = dh - v dp$$

$$du + p dv = c_v dT + p dv \quad ; \quad dh - v dp = c_p dT - v dp$$

$$dw_v = c_v \frac{k-1}{n-1} dT = -p dv$$

For the change in entropy of the polytropic state change:

$$ds = c_v \frac{n-k}{n-1} \frac{dT}{T}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{n-1} = \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}}$$

$$s_2 - s_1 = c_v \frac{n-k}{n-1} \ln \frac{T_2}{T_1} \quad n \neq 1$$

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R_i \ln \frac{v_2}{v_1} = c_p \ln \frac{T_2}{T_1} - R_i \ln \frac{p_2}{p_1}$$

Compressor with dissipation: $n > k$

Turbine with dissipation: $n < k$

11. Exergy and anergy

Exergy: useful part of energy that can be completely converted into work under given ambient conditions.

Anergy: remaining part of energy that cannot be converted into work under given ambient conditions.

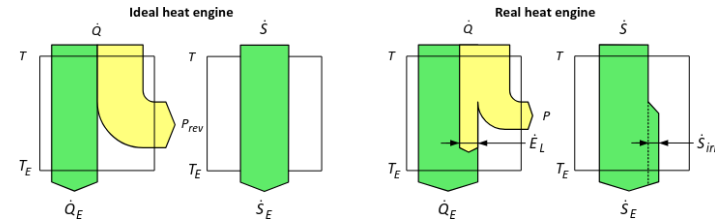
$$e = ex + an$$

11.1 Statements of the 2nd LT

- I. In all irreversible processes exergy is converted in anergy
- II. Only in reversible processes exergy remains constant
- III. It is impossible to convert anergy into exergy

11.2 Exergy

11.2.1 Exergy balance / Exergy loss



$$\dot{E}_L = \sum_{\alpha} \dot{E}_{\alpha} - \sum_{\omega} |\dot{E}_{\omega}| = \dot{Q} \frac{T_A - T_E}{T_A} + \left(-\dot{Q} \frac{T_B - T_E}{T_B} \right)$$

$$\dot{E}_L = T_E \dot{S}_{\text{irr}} = T_E \frac{T_A - T_B}{T_A T_B} \dot{Q}$$

$$\dot{S}_{\text{irr}} = \sum_{\omega} \dot{m}_{\omega} s_{\omega} - \sum_{\alpha} \dot{m}_{\alpha} s_{\alpha} - \sum \dot{S}_Q \geq 0, \quad d\dot{S}_Q = \frac{d\dot{Q}}{T}$$

11.2.2 Exergy of heat

$$\dot{E}_Q = P_{\text{rev}} = \left(1 - \frac{T_E}{T} \right) \dot{Q} = \eta_C \dot{Q}$$

11.2.3 Exergy of material flows (without mixtures)

$$e = h - h_E - T_E (s - s_E) + \frac{c^2}{2} + gz + q \left(1 - \frac{T_E}{T} \right)$$

$$e_2 - e_1 = h_2 - h_1 - T_E (s_2 - s_1)$$

Perfect gases

$$e = c_p \left(T - T_E - T_E \ln \frac{T}{T_E} \right) + R_i T_E \ln \frac{p}{p_E}$$

Incompressible liquids (isobaric)

$$e = c_p \left(T - T_E - T_E \ln \frac{T}{T_E} \right)$$

Isobaric for $T < T_E$ (cooling)

$$e = c_p \left(T_E - T - T_E \ln \frac{T_E}{T} \right)$$

III. Steady-state flow processes

12. Adiabatic working cycles

12.1 Compressor

12.1.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2} (c_2^2 - c_1^2) + g(z_2 - z_1)$$

$$w_{t12} + q_{12} = c_p (T_2 - T_1) + \frac{1}{2} (c_2^2 - c_1^2) + g(z_2 - z_1)$$

$$P_{12} + \dot{Q}_{12} = \dot{m} (w_{t12} + q_{12})$$

12.1.2 Isentropic compression efficiency factor

$$\eta_s = \frac{\text{isentropic work}}{\text{real polytropic work}}$$

$$\eta_s = \frac{\frac{\Delta h_s}{\Delta h}}{\frac{h_{2,s} - h_1}{h_2 - h_1}} = \frac{c_{p,m} (T_{2,s} - T_1)}{c_{p,m} (T_2 - T_1)} = \frac{T_{2,s} - T_1}{T_2 - T_1} \leq 1$$

12.2 Turbine

12.2.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2} (c_2^2 - c_1^2) + g(z_2 - z_1)$$

$$w_{t12} + q_{12} = c_p (T_2 - T_1) + \frac{1}{2} (c_2^2 - c_1^2) + g(z_2 - z_1)$$

$$P_{12} = \dot{m} w_{t12} \iff -P_{12} = \dot{m} |w_{t12}|$$

12.2.2 Isentropic turbine efficiency

$$\eta_s = \frac{\text{real polytropic work}}{\text{isentropic work}}$$

$$\eta_s = \frac{-\Delta h}{-\Delta h_s} = \frac{h_1 - h_2}{h_1 - h_{2,s}} = \frac{T_1 - T_2}{T_1 - T_{2,s}}$$

13. Adiabatic flow processes without work

13.1 Throttle / Expansion valve (isenthalpic)

$$h_1 = h_2 \implies j_{12} = - \int_1^2 v dp = -R_i T \ln \frac{p_2}{p_1}$$

13.2 Nozzle

$$h_2^0 = h_1^0 \implies h_2 + \frac{1}{2} c_2^2 = h_1 + \frac{1}{2} c_1^2$$

$$c_2 = \sqrt{2(h_1 - h_2) + c_1^2}$$

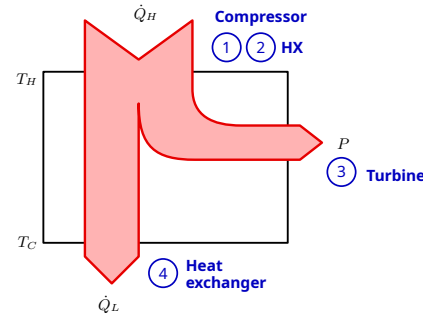
where h^0 is the total enthalpy: $h^0 = h + \frac{c^2}{2}$

IV. Thermodynamics cycles

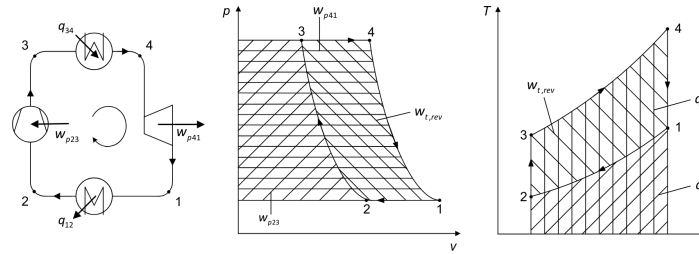
14. Reversible cycle work

Total cycle / sum of all sub-processes:

$$w_{t,rev} = \int_1^2 v dp + \int_2^3 v dp + \dots + \int_n^1 v dp = \oint v dv = \sum_{i=1}^n w_{pi}$$



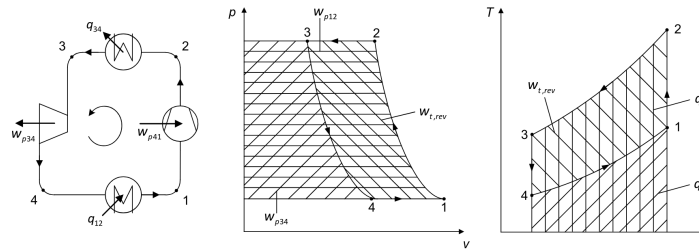
14.1 Clockwise cycle



$$w_{t,rev} = w_{p,23} + w_{p,41} < 0$$

$$|w_{t,rev}| = |q_{34}| - |q_{12}|$$

14.2 Counterclockwise cycle

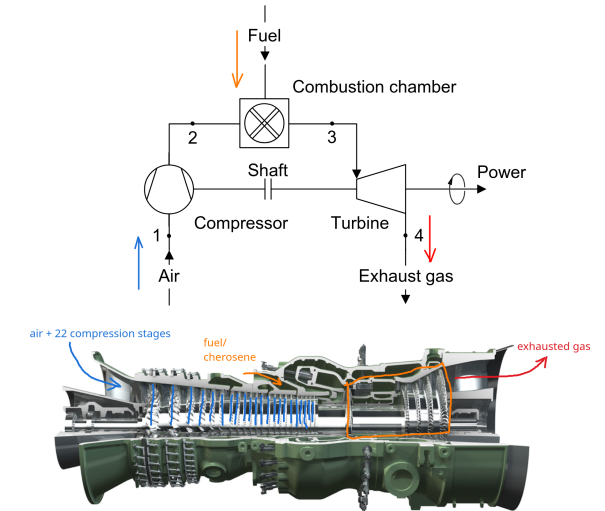


$$w_{t,rev} = w_{p,12} + w_{p,34} < 0$$

$$|w_{t,rev}| = |q_{23}| - |q_{41}|$$

V. Thermal systems with gas turbines

15. Turbine



Power balance:

$$\dot{Q}_{23} + \dot{Q}_{41} + P_C = 0$$

15.1 Joule process

$$\pi = \frac{p_2}{p_1} = \frac{p_3}{p_4}$$

$$-w_C = q_\alpha + q_\omega = q_{23} + q_{41} = c_p (T_3 - T_2 + T_1 - T_4)$$

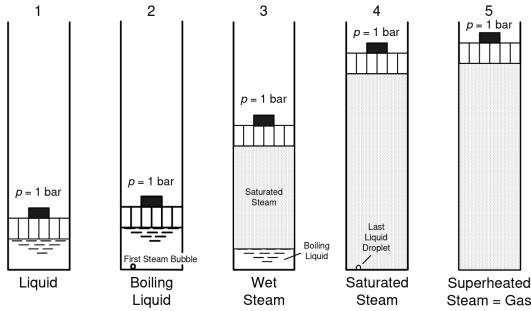
$$|w_C| = c_p (T_3 - T_2) \left(1 - \frac{T_4 - T_1}{T_3 - T_2} \right)$$

$$\left(1 - \frac{T_4 - T_1}{T_3 - T_2} \right) = \left(\frac{p}{p_0} \right)^{\frac{k-1}{k}} = \left(\frac{1}{\pi} \right)^{\frac{k-1}{k}}$$

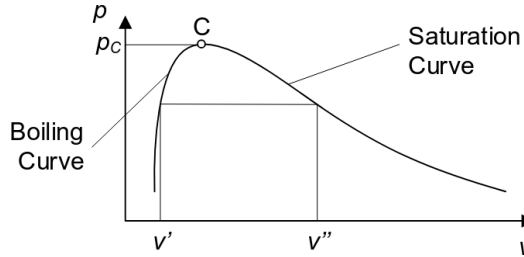
$$\eta_J = \frac{|w_C|}{q_\alpha} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{1}{\pi} \right)^{\frac{k-1}{k}}$$

VI. Thermal plants with steam turbines

16. Types of steam



17. Boundary curves of the two phase region



17.1 Steam tables

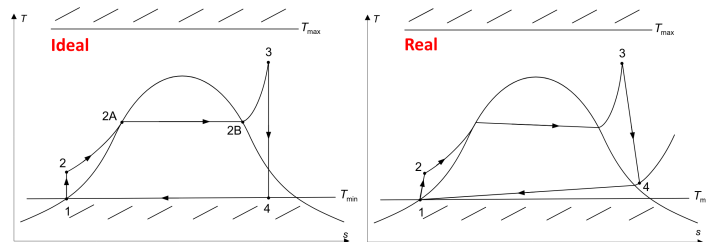
$$v = (1 - x)v' + xv'' = v' + x(v'' - v')$$

$$h = (1 - x)h' + xh'' = h' + x\Delta h_v$$

$$s = (1 - x)s' + xs'' = s' + x\left(\frac{\Delta h_v}{T}\right)$$

18. Steam power stations

18.1 Rankine cycle



$$w_{t12} = h_2 - h_1 \text{ (pump)}$$

$$q_{23} = h_3 - h_2 \text{ (boiler)}$$

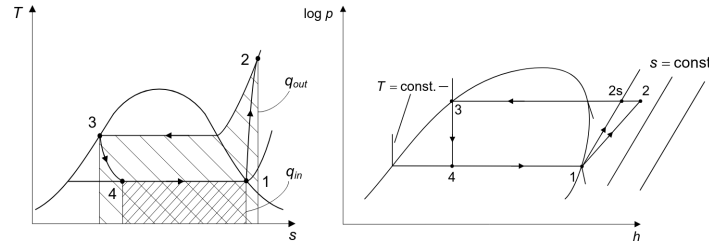
$$|w_{t34}| = h_3 - h_4 \text{ (turbine)}$$

$$|q_{41}| = h_4 - h_1 \text{ (condensor)}$$

$$\eta_{th} = \frac{(h_3 - h_4) - \overbrace{(h_2 - h_1)}^{\text{pump work}}}{h_3 - h_2} ; \quad \eta_s = \frac{h_3 - h_4}{h_3 - h_{4,s}} \leq 1$$

VII. Heat pumps and Chillers

19. T-s and log p-h diagrams



$$w_{t,12} = h_2 - h_1 \text{ (compressor)}$$

$$q_{23} = h_2 - h_3 \text{ (condenser)}$$

$$h_3 = h_4 \text{ (expansion valve)}$$

$$q_{41} = h_1 - h_4 \text{ (evaporator)}$$

19.1 Isentropic efficiency (compressor)

$$\eta_s = \frac{h_{2,s} - h_1}{h_2 - h_1} \leq 1$$

20. Coefficient of performance

20.1 Inner coefficient of performance

20.1.1 Inner COP of a heat pump

$$\varepsilon_{HP} = \frac{|q_{23}|}{w_{12}} = \frac{\text{heat}}{\text{technical work}} = \frac{h_2 - h_3}{h_2 - h_1} = \varepsilon_R + 1$$

$$\text{Carnot: } \frac{1}{\eta_C} = \varepsilon_{HP,C} = \frac{T_C}{T_C - T_E} = \frac{T_C}{\Delta T_{\text{lift}}} = \varepsilon_{R,C} + 1$$

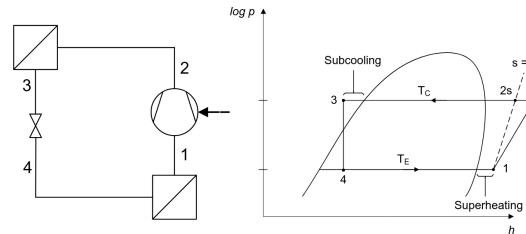
20.1.2 Inner COP of a refrigerator

$$\varepsilon_R = \frac{|q_{41}|}{w_{12}} = \frac{h_1 - h_4}{h_2 - h_1} = \varepsilon_{HP} - 1$$

$$\text{Carnot: } \varepsilon_{R,C} = \frac{T_E}{T_C - T_E} = \frac{T_E}{\Delta T_{\text{lift}}} = \varepsilon_{HP,C} - 1$$

20.2 Outer coefficient of performance (COP)

20.2.1 COP of a heat pump



$$\text{COP}_{HP} = \frac{|\dot{Q}_w|}{P_{el}} = \frac{|\dot{Q}|}{P_{el}}$$

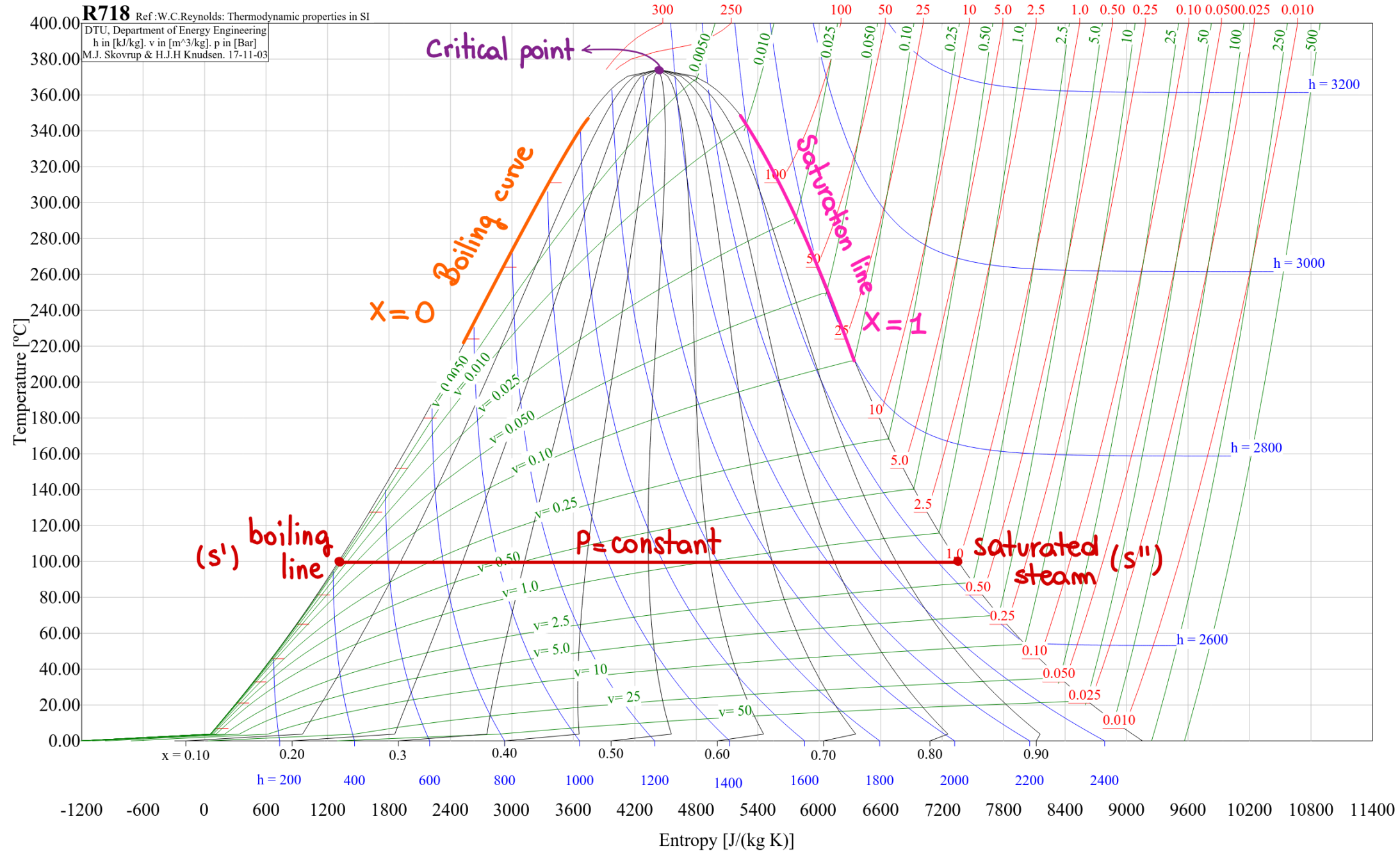
$$\varepsilon_{HP} > \text{COP}_{HP} ; \quad \zeta_{HP} = \frac{\text{COP}_{HP}}{\varepsilon_{HP,C}}$$

20.2.2 COP of a refrigerator

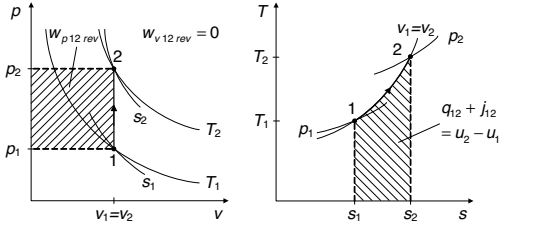
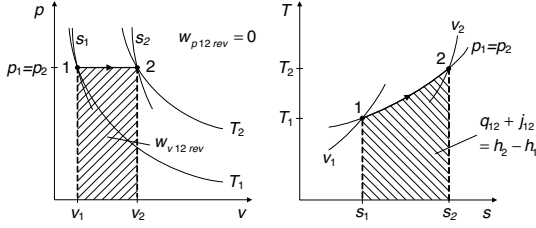
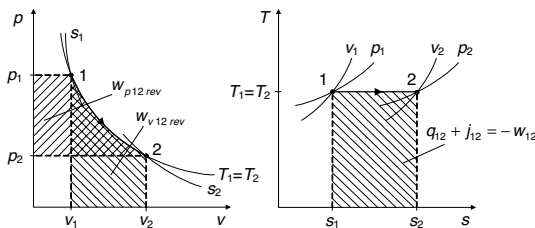
$$\text{EER} = \text{COP}_R = \frac{\dot{Q}_{in}}{P_{el}} = \frac{\dot{Q}_e}{P_{el}}$$

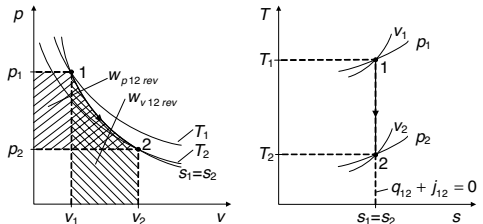
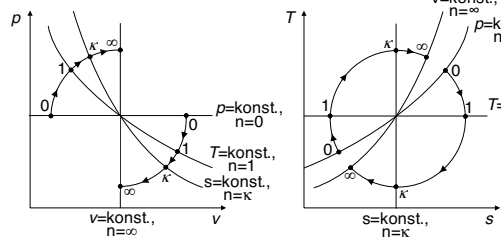
$$\varepsilon_R > \text{EER} ; \quad \zeta_R = \frac{\text{EER}}{\varepsilon_{R,C}}$$

A. T-s diagram



Special state changes with perfect gases (without consideration of kinetic and potential energies)

Ideal gas	$v = \text{constant}$: Isochoric state change	$p = \text{constant}$: Isobaric state change	$T = \text{constant}$: Isothermal state change
Therm. state equation $pV = nRT$ $R = 8.314 \text{ J/mol K}$ $pV = mR_i T$ $R_i = \frac{R}{M_i}$ $pv = R_i T, \quad \rho = \frac{p}{R_i T}$ $R_i = c_p - c_v, \quad \kappa = \frac{c_p}{c_v}$ $c_v = \frac{R_i}{\kappa - 1}, \quad c_p = \frac{\kappa R_i}{\kappa - 1}$	 $v = \text{const.}: \quad \frac{p_1}{T_1} = \frac{p_2}{T_2} = \frac{p}{T} = \text{const.}$ $w_{v12rev} = 0 \quad (dv = 0)$ $w_{p12rev} = v(p_2 - p_1) = \frac{p_2 - p_1}{\rho}$	 $p = \text{const.}: \quad \frac{v_1}{T_1} = \frac{v_2}{T_2} = \frac{v}{T} = \text{const.}$ $w_{v12rev} = -p(v_2 - v_1) = -R_i(T_2 - T_1)$ $w_{p12rev} = 0 \quad (dp = 0)$	 $T = \text{const.}: \quad p_1 v_1 = p_2 v_2 = pv = \text{const.}$ $w_{v12rev} = -p_1 v_1 \ln \frac{v_2}{v_1} = p_1 v_1 \ln \frac{p_2}{p_1} = R_i T \ln \frac{p_2}{p_1}$ $w_{p12rev} = w_{v12rev}$
1st LT closed system $q_{12} + w_{12} = u_2 - u_1$ Rev. volume change work: $w_{v12rev} = -\int_1^2 p dv$ $w_{v12} = w_{v12rev} + j_{12}$ Overall work: $w_{12} = w_{v12} + w_{W12} + w_{el12}$	$q_{12} + w_{12} = u_2 - u_1$ Rev. SC: $w_{12} = w_{v12rev} = 0 \quad (dv = 0)$ $q_{12rev} = u_2 - u_1 = c_v(T_2 - T_1)$	$q_{12} + w_{12} = u_2 - u_1$ Rev. SC: $w_{12} = w_{v12rev} = -p(v_2 - v_1)$ $q_{12rev} = u_2 - u_1 + p(v_2 - v_1) = h_2 - h_1$ $q_{12rev} = c_p(T_2 - T_1)$	$q_{12} + w_{12} = u_2 - u_1 = c_v(T_2 - T_1) = 0$ $q_{12} = -w_{12}$ Rev. SC: $q_{12rev} = -w_{v12rev}$ $q_{12rev} = p_1 v_1 \ln \frac{v_2}{v_1} = -p_1 v_1 \ln \frac{p_2}{p_1} = -R_i T \ln \frac{p_2}{p_1}$
1st LT open system $q_{12} + w_{t12} = h_2 - h_1$ Rev. pressure change work: $w_{p12rev} = \int_1^2 v dp$ Technical work: $w_{t12} = \int_1^2 v dp + j_{12}$	$q_{12} + w_{t12} = h_2 - h_1$ Rev. SC: $w_{t12} = w_{p12rev} = v(p_2 - p_1)$ $q_{12rev} = h_2 - h_1 - v(p_2 - p_1) = u_2 - u_1$ $q_{12rev} = c_v(T_2 - T_1)$	$q_{12} + w_{t12} = h_2 - h_1$ Rev. SC: $w_{t12} = w_{p12rev} = 0 \quad (dp = 0)$ $q_{12rev} = h_2 - h_1 = c_p(T_2 - T_1)$	$q_{12} + w_{t12} = h_2 - h_1 = c_p(T_2 - T_1) = 0$ $q_{12} = -w_{t12}$ Rev. SC: $q_{12rev} = -w_{p12rev}$ $q_{12rev} = p_1 v_1 \ln \frac{v_2}{v_1} = -p_1 v_1 \ln \frac{p_2}{p_1} = -R_i T \ln \frac{p_2}{p_1}$
2nd LT / entropy $ds = \frac{dq + dj}{T}$ $ds = \frac{du + p dv}{T} = \frac{dh - v dp}{T}$	$ds = \frac{du}{T} = \frac{c_v dT}{T} \Rightarrow s_2 - s_1 = c_v \ln \frac{T_2}{T_1}$	$ds = \frac{dh}{T} = \frac{c_p dT}{T} \Rightarrow s_2 - s_1 = c_p \ln \frac{T_2}{T_1}$	$ds = \frac{p dv}{T} = \frac{R_i dv}{v} \Rightarrow s_2 - s_1 = R_i \ln \frac{v_2}{v_1}$ $ds = -\frac{v dp}{T} = -\frac{R_i dp}{p} \Rightarrow s_2 - s_1 = -R_i \ln \frac{p_2}{p_1}$

Ideal gas	$s = \text{constant}$: Isentropic state change	$n = \text{constant}$: Polytropic state change
Therm. state change $pV = nRT$ $R = 8.314 \text{ J/mol K}$ $pV = mR_iT$ $R_i = \frac{R}{M_i}$ $pv = R_iT, \quad \rho = \frac{p}{R_iT}$ $R_i = c_p - c_v, \quad \kappa = \frac{c_p}{c_v}$ $c_v = \frac{R_i}{\kappa - 1}, \quad c_p = \frac{\kappa R_i}{\kappa - 1}$	 $s = \text{const} : \quad p_1 v_1^\kappa = p_2 v_2^\kappa = \text{const.} \quad \Rightarrow \quad \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} = \left(\frac{v_1}{v_2}\right)^{\kappa-1}$ $w_{v12rev} = \frac{p_1 v_1}{\kappa - 1} \left[\left(\frac{v_1}{v_2}\right)^{\kappa-1} - 1 \right] = \text{etc. (replace)}$ $w_{p12rev} = \kappa w_{v12rev} = \kappa \frac{p_1 v_1}{\kappa - 1} \left[\left(\frac{v_1}{v_2}\right)^{\kappa-1} - 1 \right] = \text{etc. (replace)}$	 Polytrope: $p v^n = \text{konst.}$ Isobare: $n=0, p v^0 = \text{konst.}$ Isotherme: $n=1, p v^1 = \text{konst.}$ Isentropie: $n=\kappa, p v^\kappa = \text{konst.}$ Isochore: $n=\infty, p v^\infty = \text{konst.}$ $n = \text{const} : \quad p_1 v_1^n = p_2 v_2^n = \text{const.} \quad \Rightarrow \quad \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} = \left(\frac{v_1}{v_2}\right)^{n-1}$ $w_{v12} = \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc. (replace, } n \neq 1)$ $w_{p12} = n w_{v12} = n \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc. (replace, } n \neq 1)$
1st LT closed system $q_{12} + w_{12} = u_2 - u_1$ Rev. volume change work: $w_{v12rev} = - \int_1^2 p dv$ $w_{v12} = w_{v12rev} + j_{12}$ Overall work: $w_{12} = w_{v12} + w_{W12} + w_{el12}$	$q_{12} + w_{12} = u_2 - u_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{12} = w_{v12rev} = u_2 - u_1 = c_v (T_2 - T_1)$ $w_{v12rev} = \frac{p_1 v_1}{\kappa - 1} \left[\left(\frac{v_1}{v_2}\right)^{\kappa-1} - 1 \right] = \frac{R_i T_1}{\kappa - 1} \left[\left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1 \right] = \text{etc.}$	$q_{12} + w_{12} = u_2 - u_1 = c_v (T_2 - T_1)$ $w_{12} = w_{v12} = \frac{R_i}{n - 1} (T_2 - T_1) = c_v \frac{\kappa - 1}{n - 1} (T_2 - T_1) =$ $= \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{12} = w_{v12rev}$ $q_{12rev} = c_v \frac{n - \kappa}{n - 1} (T_2 - T_1) = \frac{n - \kappa}{\kappa - 1} \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc.} \quad (n \neq 1)$
1st LT open system $q_{12} + w_{t12} = h_2 - h_1$ Rev. pressure change work: $w_{p12rev} = \int_1^2 v dp$ Technical work: $w_{t12} = \int_1^2 v dp + j_{12}$	$q_{12} + w_{t12} = h_2 - h_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{t12} = w_{p12rev} = \kappa w_{v12rev} = h_2 - h_1 = c_p (T_2 - T_1)$ $w_{p12rev} = \kappa \frac{p_1 v_1}{\kappa - 1} \left[\left(\frac{v_1}{v_2}\right)^{\kappa-1} - 1 \right] = \kappa \frac{R_i T_1}{\kappa - 1} \left[\left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1 \right] = \text{etc.}$	$q_{12} + w_{t12} = h_2 - h_1 = c_p (T_2 - T_1)$ $w_{t12} = w_{p12} = n w_{v12} = n \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{p12rev} = n w_{v12rev} = \frac{n R_i}{n - 1} (T_2 - T_1) = \text{etc.}$ $q_{12rev} = c_v \frac{n - \kappa}{n - 1} (T_2 - T_1) = \frac{n - \kappa}{\kappa - 1} \frac{p_1 v_1}{n - 1} \left[\left(\frac{v_1}{v_2}\right)^{n-1} - 1 \right] = \text{etc.} \quad (n \neq 1)$
2nd LT / entropy $ds = \frac{dq + dj}{T}$ $ds = \frac{du + p dv}{T} = \frac{dh - v dp}{T}$	$ds = \frac{dq + dj}{T} = 0 \Rightarrow s_2 - s_1 = 0$	$s_2 - s_1 = c_v \frac{n - \kappa}{n - 1} \ln \frac{T_2}{T_1} \quad (\text{for } n \neq 1)$ $s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R_i \ln \frac{v_2}{v_1} = c_p \ln \frac{T_2}{T_1} - R_i \ln \frac{p_2}{p_1}$